

Steven Hoekstra

PhD Candidate in Economics, University of Amsterdam & Tinbergen Institute

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Profile

My research is focussed on macroeconomics viewed through the lens of complex systems. I have gained experience in both equilibrium and agent-based modelling through my academic pursuits. My objective is to use insights of both subfields to find a better understanding of macroeconomic phenomena using theory, large scale computational modelling, and empirical methods.

Interests

Complexity, macroeconomics, bounded rationality, agent-based modelling, networks, distributed computing, heterogeneous agent equilibrium models.

Education

PhD in Economics — University of Amsterdam 2023 — current

Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers

MPhil Economics and Econometrics — Tinbergen Institute (*cum laude*) 2021 — 2023

Specialisation: Complexity, Mathematics, and Computation

Thesis: An Agent Based Model of the Dutch Economy

Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers

MSc Econometrics — University of Amsterdam (*cum laude*) 2020 — 2021

Specialisation: Complexity and Economic Behaviour

Thesis: Endogenous Financial Instability via Heterogeneous Beliefs

Supervisor: Prof. dr. C.H. Hommes

MSc Economics — London School of Economics 2019 — 2020

Specialisation: Public Economics

Thesis: The Inequality of Education using a Two Agent New Keynesian Model

Supervisor: Prof. dr. F. Cowell

Exchange Programme — Royal Melbourne Institute of Technology 2018 — 2019

Minor in Finance and Programming

BSc Economics and Business Economics — University of Groningen 2016 — 2019

Thesis: The Impact of News Coverage on Security Prices

BSc Econometrics — University of Groningen 2015 — 2018

Thesis: Arbitrage Smoothing and Term Structure Models

Working Papers

Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households

with Cars Hommes and Frank Pijpers

Finalist, Society for Computational Economics student paper contest, CEF 2026

BeforeIT.jl: High-Performance Agent-Based Macroeconomics Made Easy

with Adriano Meligrana, Mitja Devetak, Sebastian Poledna, Oliver Reiter and Aldo Glielmo

Work in Progress

Estimation, Forecasting and Parameter Constancy in a Dutch Agent-Based Model An Agent-Based Model of the Housing Market

Presentations

2026	Computing in Economics and Finance (CEF), Society for Computational Economics
2026	Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA)
2025	Agent-Based Modelling for Policy (ABM4Policy), European Central Bank, Frankfurt
2025	7th Behavioral Macroeconomics Workshop, University of Bamberg
2025	MacroABM 2nd Workshop, <i>Microdata in Macromodels</i> , Vienna University of Economics and Business
2025	Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA)
2025	De Nederlandsche Bank, Amsterdam
2023	De Nederlandsche Bank, Amsterdam

Work Experience

De Nederlandsche Bank — PhD Intern Apr. — Dec. 2025

- Implemented the Bank of Canada CANVAS agent-based model for the Netherlands
- Estimated inflation channels using neural ratio estimation
- Wrote a note on the wage-price channel using this estimated model

Statistics Netherlands (CBS) — Thesis Intern Mar. — Sept. 2023

- Calibrated the agent-based model of Poledna et al. (2023) with Dutch register and national accounts data
- Developed an extension incorporating optimal consumption
- Ran simulations with both models and analysed their internal dynamics

SEO Amsterdam Economics — Research Assistant 2020 — 2021

- Researched academic pressure among secondary school students
- Conducted stakeholder interviews and set up a nationwide survey
- Performed statistical analysis on the survey outcomes

Teaching

Graduate

Behavioural Macro and Finance — Lecturer 2025

Undergraduate

Mathematics 3: Advanced Linear Algebra — Tutor 2025 — 2026

Mathematics 2: Linear Algebra — Tutor 2024 — 2025

Microeconomics for AE — Tutor 2023 — 2024

Achievements

Honours

Cum Laude — MPhil Economics and Econometrics, Tinbergen Institute 2023

Cum Laude — MSc Econometrics, University of Amsterdam 2021

Scholarships

Full scholarship (merit based) — Tinbergen Institute 2022

Tuition fee waiver — Tinbergen Institute 2021 — 2023

Visits and Summer Schools

Visiting Scholar — IIASA, Laxenburg Aug. 2023

International Institute for Applied Systems Analysis. Visited Dr. S. Poledna to work on agent-based models

Tools for Macroeconomists: Advanced Tools — University of Oxford Aug. 2021

Doctoral-level course on DSGE models: solving and simulating models with heterogeneous agents, continuous time models, occasionally binding constraints and bounded rationality

Tools for Macroeconomists: The Essentials — University of Oxford Aug. 2021

Doctoral-level course on DSGE models: numerical analysis, perturbation and projection methods, Blanchard-Kahn conditions and sunspot solutions, the Kalman filter and Bayesian estimation

Skills

Programming	Matlab, Python, R, Julia, C, SQL, Bash
High-performance computing	Parallel computing (MPI, BSPlib)
Methods	Agent-based modelling, network methods, machine learning, econometrics, dynamic stochastic general equilibrium modelling
Packages	Dynare, Stata

References

Prof. dr. C.H. Hommes

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Prof. dr. F.P. Pijpers

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